

Recurrence Is Not Enough: Causally Validating Multilingual SAE Translation Features in Gemma 2 and 3

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Abstract

Sparse autoencoder (SAE) features are increasingly used to explain and steer language-model behavior, but it remains unclear whether a feature found in one language context plays the same causal role when processing prompts in another language. We study this question using translation-initiation features (Wu et al., 2026). We reproduce the SAE feature discovery method from Wu et al. in Gemma 2 and extend it to multilingual settings that vary prompt language, source language, and target language. We then test whether features that recur across settings affect translation behavior by amplifying or ablating their activations during inference. We also examine whether the method can be applied to Gemma 3.

In both models, we observe an identical finding: although we can find more than 20 features that activate frequently across all discovery settings, causal validation shows that nearly all have small or inconsistent effects. In contrast, one feature – Gemma 2’s (L10, 5717) and Gemma 3’s (L20, 2456) – consistently improves COMET scores when amplified and degrades them when ablated across 23 language settings. These results show that feature recurrence can overstate cross-lingual transfer, while identifying a language-agnostic translation-initiation direction in Gemma 2 and Gemma 3.¹

1 Introduction

Mechanistic interpretability seeks internal variables that causally support model behavior (Elhage et al., 2022; Olah et al., 2020). Sparse autoencoders (SAEs) are widely used to decompose activations into interpretable feature directions (Bricken et al., 2023; Huben et al., 2024). However, SAE explanations can be brittle: automated interpretability scores may fail on basic sanity checks (Heap et al.,

2026), and semantic-looking features may reflect token artifacts (Ronge et al., 2026). These concerns are sharper in multilingual settings, where a feature found in one language context may not implement the same computation elsewhere.

We study this issue through the translation-initiation mechanism of Wu et al. (2026). They report that a small set of Gemma SAE features activate at translation-specific prompt positions, and that amplifying or ablating them improves or hurts translation quality. However, their evaluation is English-centric: prompts are in English and translation is English-to-X. It is unclear whether the same directions are causal when the prompt, source, or target language changes. This question connects to broader work on whether multilingual LLMs rely on shared language-agnostic or language-specific representations (Wendler et al., 2024; Brinkmann et al., 2025; Schut et al., 2025). Moreover, their results were only reported in one model family (Gemma 2’s 2B and 9B variants), raising questions of whether the method can be transferred to other models.

In this reproduction study, we ask: *do SAE translation features causally transfer across multilingual settings?* We extend the discovery and intervention pipeline of Wu et al. (2026). We use the same models – Gemma 2 2B IT (Gemma Team, 2024) and pre-trained SAEs from the Gemma Scope suite (Lieberum et al., 2024) – but vary both translation pair and prompt language. We first identify features shared across four settings (English-to-Chinese with English prompts, Chinese-to-English with Chinese prompts, and Chinese-to-Japanese with either Chinese or English prompts) then intervene on these features in 23 settings. We deliberately remove English’s influence by introducing settings where English is not present in the source sentence, or the prompt instruction, or simply removed entirely. We evaluate effects of feature intervention using COMET scores (Rei et al., 2020).

¹Our code is available at: <https://github.com/giangson19/recurrence-is-not-enough>

Aspect	Wu et al. (2026)	Ours
Models	Gemma 2	Gemma 2, 3
Discovery settings	en2zh_en	en2zh_en, zh2en_zh, zh2ja_zh, zh2ja_en
Test settings	4; English source/prompt	23; non-English sources and no-English settings

Table 1: Comparison of experimental scope.

Our main finding is unexpectedly sparse. Although we found 22 recurring features across all four discovery settings, **only one—(L10, 5717)—shows robust causal transfer**. Amplifying it promotes translation behavior across diverse language pairs and prompt languages, while ablating it usually degrades quality. This suggests that one SAE direction functions as a language-agnostic translation-initiation switch, while the failure of most overlapping candidates cautions against treating feature recurrence as evidence of mechanism transfer.

Furthermore, we test whether the discovery pipeline can be applied to another model, specifically using Gemma 3 4B IT (Gemma Team, 2025) and pre-trained SAEs from Gemma Scope 2 (McDougall et al., 2025) of similar sizes to Gemma 2’s. Despite differences in model size, architecture and pre-training data to Gemma 2, we observe identical results in Gemma 3: we found 22 recurring features across four discovery settings, yet **only one – (L20, 2456) – consistently has improving/degrading effects** under intervention across 23 language settings.²

Our contributions are:

- We test cross-lingual transferability of SAE-based translation features across prompt languages and translation pairs.
- We show that multilingual feature recurrence overestimates causal transferability.
- We identify one feature in each model, (L10, 5717) in Gemma 2 and (L20, 2456) in Gemma 3, whose causal effects transfer robustly across settings.

The scope of our reproduction study compared to Wu et al. (2026) is summarized in Table 1.

²The identical count of 22 recurring features in both models is coincidental; it arises from independent runs of the same pipeline on different models and SAEs.

2 Related Work

Sparse dictionary learning for interpretability and intervention. SAEs decompose model activations into sparse feature activations and decoder directions, often yielding more human-interpretable units than raw neurons (Bricken et al., 2023; Huben et al., 2024; Templeton et al., 2024). Recent work uses SAEs to study multilingual mechanisms, language-specific features, and feature steering (Cho and Hockenmaier, 2025; Deng et al., 2025; Chou et al., 2025). Our work follows this line but emphasizes causal validation under distribution shift: a feature is only considered transferable if the same intervention direction changes behavior across many languages.

Reliability of mechanistic explanations. A growing literature questions whether interpretability methods identify stable mechanisms or produce plausible but brittle descriptions. Prior work highlights interpretability illusions (Bolukbasi et al., 2021), weak automated explanation metrics (Heap et al., 2026), and token-driven false positives (Ronge et al., 2026). Our results reinforce these concerns: 22 features pass a strict discovery criterion, but only one shows robust causal effects.

Multilingual representations and interpretability. Mechanistic interpretability remains largely English-dominant, and relatively little work tests whether interpretability findings retain their meaning or causal role across languages (Resck et al., 2025). This matters because multilingual models may share representations across languages, but can also exhibit language-specific computation or latent-language effects (Wendler et al., 2024; Brinkmann et al., 2025; Schut et al., 2025). Some behavioral control directions, such as refusal, appear to transfer across languages (Arditi et al., 2024; Wang et al., 2026). We build on this line by testing whether a translation-initiation feature remains causal when the prompt, source, and target language change.

3 Method

3.1 Notation

We denote the features as (Layer, Feature Index). For example, (L10, 5717) refers to the 5717th SAE feature from the 10th layer.

Moreover, we denote each setting as src2tgt_prompt, where src is the source

Setting	Gemma 2	Gemma 3
en2zh_en	42	45
zh2en_zh	53	104
zh2ja_zh	83	206
zh2ja_en	59	65
Intersection (all 4)	22	22

Table 2: Number of consistent features discovered under each setting.

language, `tgt` is the target language, and `prompt` is the language of the translation instruction. We use the language codes `en`, `zh`, `ja`, `ar`, `ru`, and `vi` for English, Chinese, Japanese, Arabic, Russian, and Vietnamese, respectively.

3.2 Feature discovery

We follow the three-stage protocol of Wu et al. (2026) for identifying translation-initiation features and use their published codebase as the foundation for our study³. All thresholds match their published values.

Stage 1: feature discovery. For each of three prompt strategies—`prompt`, `tgt_lang`, and `tgt_idx`—we run forward passes on the 98 examples with target text appended. We extract nonzero SAE activations at the corresponding boundary token position, retain features that activate in more than 60% of example-template slots, filter by minimum mean activation greater than 10.0, and keep the top 50 features by mean activation. We then take the union across the three strategies, yielding up to 150 candidate features per setting.

The boundary token position depends on the prompt strategy: `prompt` uses the token immediately before the source text, `tgt_lang` uses the last token of the target-language name in the instruction, and `tgt_idx` uses the last token of the appended target text.

Stage 2: PCA-based consistency filtering. For each candidate feature, we ablate the feature by setting its activation to zero on each discovery example across all prompt templates. We then compute the intervention-induced delta vector, defined as the change in the hidden state at the last-token position, and apply PCA to the collected delta vectors. Following Wu et al. (2026), we retain features for which the first principal component explains at least 95% of the variance. This filters for features whose

³<https://github.com/Flamewei123/AAAI26-translation-Initiation-Features>

interventions induce a consistent residual-stream direction rather than noisy or context-dependent effects.

Stage 3: causal intervention. For each surviving feature, we evaluate interventions on the 862 held-out examples. We compare greedy baseline generations with two interventions applied at the last prompt token: `ablation`, which sets the feature activation to zero ($\alpha = 0$), and `amplification`, which doubles the feature activation ($\alpha = 2$). We report COMET changes relative to the unmodified baseline.

3.3 Cross-lingual candidate selection

We run the discovery pipeline independently under four settings that vary the prompt language, source language, and target language. Table 2 summarizes the number of consistent features discovered for each setting and model. The settings progressively eliminate English-specific factors: `zh2en_zh` reverses the translation direction while using a Chinese instruction, `zh2ja_zh` removes English entirely, and `zh2ja_en` restores English only in the instruction. Therefore, features in the four-way intersection are unlikely to depend on English prompts, English source text, or a particular translation direction.

3.4 Dataset, model, and evaluation

We use WMT24++ (Deutsch et al., 2025) and construct non-English X–Y pairs by aligning examples through their shared English sentences, yielding 960 translation examples. From 6 languages `en`, `zh`, `ja`, `ar`, `ru`, and `vi`, we construct 23 `src2tgt_prompt` settings, including the 4 settings we used for feature discovery. In all experiments, we use 98 shared examples for feature discovery and the rest for intervention evaluation.

We start with the same models used by Wu et al. (2026): Gemma 2 2B IT (Gemma Team, 2024) with Gemma Scope residual stream SAEs⁴ (Lieberum et al., 2024), which has a width of 16k features.

We further test Gemma 3 4B IT (Gemma Team, 2025) and Gemma Scope 2’s SAEs with 16k width and big L0⁵. Although pre-trained SAEs with 262k width were released for Gemma 3, we focus on the 16k variant to ensure method parity with previous published results from Wu et al. (2026). There are

⁴<https://huggingface.co/google/gemma-scope-2b-pt-res>

⁵https://huggingface.co/google/gemma-scope-2-4b-it/tree/main/resid_post_all

Feature	Amplification ΔCOMET				Ablation ΔCOMET			
	en2zh_en	zh2en_zh	zh2ja_zh	zh2ja_en	en2zh_en	zh2en_zh	zh2ja_zh	zh2ja_en
(L0, 8920)	+0.10	+0.37	-0.78	-0.14	-0.27	-0.30	+0.38	+0.84
(L1, 12054)	-0.11	+0.16	-1.03	-0.35	-0.02	-0.21	+0.21	+0.44
(L2, 15089)	+0.06	-0.29	-0.92	+0.31	+0.08	-0.20	+0.44	-0.18
(L6, 13460)	-0.72	-0.46	-0.17	-0.21	-0.16	+0.15	-0.26	+0.75
(L7, 3704)	+0.11	-0.14	-0.06	-0.32	-0.44	+0.07	-0.55	-0.09
(L8, 2024)	-0.14	-0.10	-0.62	-0.08	+0.41	+0.56	+0.63	+0.28
(L9, 54)	-0.89	-0.03	-0.71	-0.75	-0.02	+0.16	-0.32	+0.97
(L10, 3986)	-1.16	-0.50	-1.37	+0.10	+0.21	+0.51	+0.77	+0.45
(L10, 5717)	+4.61	+2.33	+5.22	+3.66	-5.85	-0.49	-3.96	-4.79
(L11, 3151)	-1.96	-0.18	-1.60	-0.09	+1.01	+0.19	+1.00	+0.24
(L12, 2620)	-1.75	-0.48	-1.68	+0.10	+0.67	+0.45	+1.35	+0.04
(L13, 11256)	-1.09	-0.90	-1.27	+0.07	+0.05	+0.10	+0.90	+0.49
(L14, 7214)	-0.67	-0.68	-0.57	+0.04	+0.07	+0.66	+0.40	+0.77
(L14, 12319)	-0.45	-0.04	+0.04	+0.81	-0.29	-0.32	+0.12	-0.08
(L15, 8610)	+0.07	-0.70	-0.71	+0.97	-0.81	+0.96	+0.12	+0.08
(L16, 3847)	-0.36	+0.45	+0.29	+0.17	-0.66	-0.77	-0.02	+0.59
(L16, 10480)	+0.28	-0.36	+0.44	+1.03	-0.69	-0.05	-0.55	-0.16
(L17, 7921)	+0.03	-0.34	-0.19	+1.57	-1.14	+0.11	-0.53	-0.55
(L18, 15394)	+0.05	-0.46	-0.22	+0.62	-0.62	+0.30	-0.23	-0.11
(L24, 8119)	+1.80	-1.13	+1.01	+0.91	-3.07	+0.84	-1.89	-1.44
(L25, 348)	-1.96	-19.72	-4.52	-8.70	-8.82	+2.96	-8.74	-12.07
(L25, 14186)	+1.67	-5.73	-3.62	-1.28	-9.12	+1.61	-5.72	-7.59

Table 3: **Gemma 2**: causal effects of the 22 features recovered in all four discovery settings. Values are changes in COMET relative to baseline; green/red cells mark effects with absolute value ≥ 1.0 . Exactly one feature, **(L10, 5717)**, shows the expected translation-switch signature (consistent positive amplification, consistent negative ablation); the remaining 21 features survived the same recurrence-based selection in the same settings, yet produce effects that are small, inconsistent in sign across settings, or both. Corresponding chrF++ results are reported in Appendix E.

also 16k SAEs with small L0, but in our preliminary testing, this variant yielded only four PCA-consistent features, which is an order of magnitude lower than with big L0. This result might be due to the lower number of activating features.

Translation quality is measured with Unbabel/wmt22-comet-da (Rei et al., 2020).

4 Results

4.1 Most overlapping features do not causally transfer

Multilingual feature recurrence does not imply causal transfer. Tables 3 and 4 show the intervention effects of all 22 shared candidates in the four discovery settings, for Gemma 2 and Gemma 3 respectively. In both models, the 21 features other than the single exception form a matched null: they passed the same PCA Consistency filter in the same settings, yet their interventions yield effects that are small, inconsistent in sign across settings, or even opposite to the expected translation-switch direction. Critically, these features lack the expected switch signature (positive amplification and negative ablation at comparable magnitude), even in their own discovery setting. Thus, passing PCA

Consistency does not isolate features with large behavioral effects, even when the language configuration is held fixed.

Only one shared feature per model has the expected causal signature. In Gemma 2, (L10, 5717) amplification improves COMET in all four discovery settings, with gains between +2.33 and +5.22, while ablation substantially hurts three of them. In Gemma 3, (L20, 2456) amplification improves COMET in all four settings, with gains up to +2.23, while ablation hurts all four, between -3.29 and -4.15. In each model, no other feature produces effects of comparable magnitude in both directions. The combination of consistent sign and magnitude, and separation from the matched null is what distinguishes a transferable causal direction from a recurrence artifact.

Appendix C reports a second control for Gemma 2: 20 features that were consistent in en2zh_en but absent from the four-way intersection produce maximum amplification effects of only +0.65 ΔCOMET , indicating that the intersection did not miss another strong translation-initiation feature.

Feature	Amplification Δ COMET				Ablation Δ COMET			
	en2zh_en	zh2en_zh	zh2ja_zh	zh2ja_en	en2zh_en	zh2en_zh	zh2ja_zh	zh2ja_en
(L4, 2491)	-0.21	-0.40	+0.13	-0.83	+0.67	+0.21	+0.58	+0.86
(L6, 40)	+0.29	-0.02	+0.05	+0.18	+0.09	-0.08	+0.08	+0.01
(L6, 236)	+0.22	-0.07	+0.42	+0.15	-0.17	-0.14	+0.37	+0.15
(L11, 6005)	+0.54	+0.22	+0.73	+0.18	-0.16	-0.26	+0.05	-0.11
(L12, 273)	-3.14	-1.39	-2.05	-6.41	-19.72	-10.49	-17.94	-18.67
(L16, 1562)	+0.40	+0.22	+0.38	-0.29	+0.17	-0.22	+0.23	-0.40
(L16, 4659)	-0.05	-0.06	+0.08	-0.42	+0.16	+0.07	+0.77	+0.08
(L17, 1125)	+0.04	-0.69	-0.46	-0.53	-4.05	-1.07	-2.57	-3.18
(L18, 11066)	+0.58	+0.72	+0.95	+0.79	-0.62	-1.30	-0.46	-1.95
(L19, 383)	+0.10	+0.66	+1.31	+0.39	-1.12	-1.06	-0.89	-2.16
(L20, 189)	+0.76	+0.67	+1.18	+0.65	-0.43	-0.71	+0.12	-0.79
(L20, 2456)	+1.40	+1.45	+2.23	+0.59	-3.29	-3.49	-3.78	-4.15
(L21, 862)	+0.77	+0.38	+0.82	+0.31	-1.04	-0.51	-0.19	-1.45
(L22, 373)	+0.63	+0.33	+0.46	+0.47	-0.03	-0.55	+0.37	-0.05
(L22, 455)	+0.48	+0.30	+0.52	+0.54	-0.16	-0.32	+0.01	-0.23
(L23, 455)	+0.61	+0.32	+0.95	+0.52	-0.42	-0.50	-0.21	-0.57
(L24, 168)	+0.31	+0.48	+0.96	+0.62	-0.66	-3.02	-0.07	-0.91
(L24, 297)	+0.36	+0.43	+1.41	-0.73	-0.48	-0.53	-1.06	-1.04
(L25, 125)	+1.01	+0.77	+0.71	+0.64	-0.58	-0.89	+0.10	-0.89
(L25, 278)	+0.82	+0.77	+1.31	+0.76	-0.90	-1.45	+0.07	-1.20
(L30, 2639)	+0.67	+0.45	+0.88	+0.35	-0.25	-0.47	-0.28	-0.89
(L33, 542)	+0.20	+0.39	+0.55	-0.11	-0.20	+0.37	+0.37	+0.11

Table 4: **Gemma 3**: causal effects of the 22 features recovered in all four discovery settings, following the same protocol and criterion as Table 3. Exactly one feature, **(L20, 2456)**, shows the expected translation-switch signature (consistent positive amplification, consistent negative ablation). Note that layer indices are not comparable across models. Corresponding chrF++ results are reported in Appendix E.

4.2 A single feature generalizes across multiple language settings

The same feature remains causal far beyond the settings used to discover it. We next intervene on the transferable feature of each model across additional prompt languages and translation pairs. Table 5 shows a broad effect in both. For Gemma 2, amplification of (L10, 5717) improves COMET in every tested setting, with gains ranging from +1.00 to +7.46. For Gemma 3, amplification of (L20, 2456) improves COMET in all settings but ar2en_ar, with significant gains of up to +2.50. Ablation degrades COMET in every setting for Gemma 2 and in all but ar2en_ar for Gemma 3, with the largest drops in en2ru, en2ja, and en2vi for Gemma 2, and in en2zh and zh2ja with non-English prompts for Gemma 3.

Amplification gains in Gemma 3 are consistently below those in Gemma 2, although they remain significant in 18 of 23 settings and ablation effects remain comparable in size. One possible reason is the higher baseline: Gemma 3 scores 8 to 18 COMET points above Gemma 2 in almost every setting, leaving less headroom for amplification gains. This is expected as Gemma 3 was trained on multilingual data with multilingual tasks (such

as translation) in mind, while Gemma 2 was not (Gemma Team, 2025, 2024).

This points to a language-agnostic translation-initiation direction. In both models the effect persists across changes in instruction, source, and target language, including fully non-English translation settings. Notably, both Gemma 2 and Gemma Scope SAEs are trained primarily on English-centric corpora, yet (L10, 5717) generalizes across multilingual prompts and translation pairs, suggesting that translation-initiation representations may emerge in a language-agnostic form even without explicitly multilingual SAE training.

4.3 Dose-response sanity check

To test whether the intervention effect depends sharply on the chosen coefficient, we vary α for Gemma 2’s (L10, 5717) and Gemma 3’s (L20, 2456) across the four discovery settings in both models. Figure 1 shows a consistent pattern: suppressing the feature ($\alpha = 0$) hurts COMET in every setting and model, while moderate amplification improves it. The default setting $\alpha = 2.0$ already produces reliable gains, and the effect typically peaks around $\alpha = 2-3$ before saturating or reversing at larger coefficients. The sign pattern is the same in Gemma 2 and Gemma 3, with larger mag-

Setting	Gemma 2 — (L10, 5717)			Gemma 3 — (L20, 2456)		
	Base.	Amp. Δ	Abl. Δ	Base.	Amp. Δ	Abl. Δ
en2zh_en	63.33	+4.61	-5.85	74.89	+1.40	-3.29
zh2en_zh	68.17	+2.33	-0.49	76.25	+1.45	-3.49
zh2ja_zh	62.66	+5.22	-3.96	76.88	+2.23	-3.78
zh2ja_en	67.34	+3.66	-4.79	77.75	+0.59	-4.15
en2ja_en	61.03	+5.59	-7.49	77.19	+1.04	-4.03
ja2zh_en	62.19	+3.07	-4.11	72.39	+1.19	-4.07
en2ar_en	51.70	+3.34	-5.25	69.55	+0.58	-1.63
en2ar_ar	55.15	+1.14	-5.23	69.18	+0.71	-1.39
ar2en_en	67.00	+1.36	-1.03	74.07	+0.14 [†]	-1.32
ar2en_ar	64.20	+1.24	-1.56	69.77	-0.41 [†]	+0.03 [†]
en2ru_en	57.39	+7.46	-8.93	74.09	+0.87	-1.64
ru2en_ru	66.92	+1.40	-1.23	73.84	+0.34 [†]	-1.80
ru2zh_en	58.49	+3.61	-4.50	70.69	+1.34	-4.18
en2zh_zh	68.61	+3.70	-5.09	76.66	+2.27	-3.29
en2zh_ja	57.67	+1.57	-2.60	65.82	+2.50	-4.58
en2zh_ru	68.60	+1.67	-2.78	75.24	+1.66	-4.70
ja2zh_ja	54.90	+1.00	-0.65	55.71	+0.71	-2.44
ja2zh_zh	57.83	+2.24	-1.12	74.56	+0.83	-3.68
ja2zh_ru	59.32	+1.32	-1.55	74.12	+0.06 [†]	-2.84
en2vi_en	63.13	+3.44	-6.79	75.54	+0.70	-1.26
en2vi_vi	66.24	+2.47	-4.31	72.51	+1.21	-2.28
en2zh_vi	65.00	+3.61	-4.66	73.38	+1.85	-4.95
zh2ja_vi	57.01	+2.54	-2.50	72.53	+2.44	-4.84

Table 5: Intervention effects for the single transferable feature in each model, (L10, 5717) in Gemma 2 and (L20, 2456) in Gemma 3, across the same 23 settings. Values are COMET scores or changes relative to baseline. All effects are significant at one-sided $p < 0.05$ by paired bootstrap over sentence-level scores (1000 resamples) except those marked [†]. Amplification improves COMET in every setting for Gemma 2, and significantly improves it in 19 of 23 settings for Gemma 3; ablation degrades translation in all settings for Gemma 2 and in 22 of 23 for Gemma 3. The feature therefore transfers across prompt languages and translation directions in both models, with ar2en_ar in Gemma 3 the sole setting showing no effect in either direction. Corresponding chrF++ results are reported in Appendix E.

nitudes in Gemma 2. This suggests that the causal effect is not an artifact of a single amplification coefficient.

4.4 Task-specificity check

To test whether Gemma 2’s (L10, 5717) improves multilingual performance in general, we apply the same amplification to non-translation tasks: XL-Sum summarization in zh/ar/ru/vi and WikiANN NER in zh/ja/ar/ru/vi. Effects are small and inconsistent: ROUGE-L changes range from -0.011 to $+0.014$, and entity-set micro-F1 changes range from -0.021 to $+0.012$. This suggests that the translation gains are task-specific rather than a generic improvement in multilingual generation or tagging. Detailed results are in Appendix D.

4.5 Feature interpretation

To quantify the impacts of feature intervention on translation initiation, we use fastText lid.176 language identifier (Joulin et al., 2017, 2016) to clas-

sify each output and label it as a non-translation when the detected language does not match the target (or the output is empty or low-confidence). Appendix B reports non-translation fractions under baseline, amplification, and ablation for both models under the 4 discovery settings. In Gemma 2, amplification reduces the non-translation rate by up to 13.7 percentage points, while ablation increases it by up to 17.3 points. Gemma 3 shows the same pattern at smaller magnitudes, consistent with its lower baseline non-translation rate. Manual inspection also supports the interpretation that (L10, 5717) controls task initiation rather than lexical choice, matching the role proposed by Wu et al. (2026).

5 Conclusion

We study whether SAE translation features retain their causal role across changes in prompt, source, and target language. We show that recurrence is a weak proxy for causal transfer. In both Gemma 2 and Gemma 3, 22 features recur across four discov-

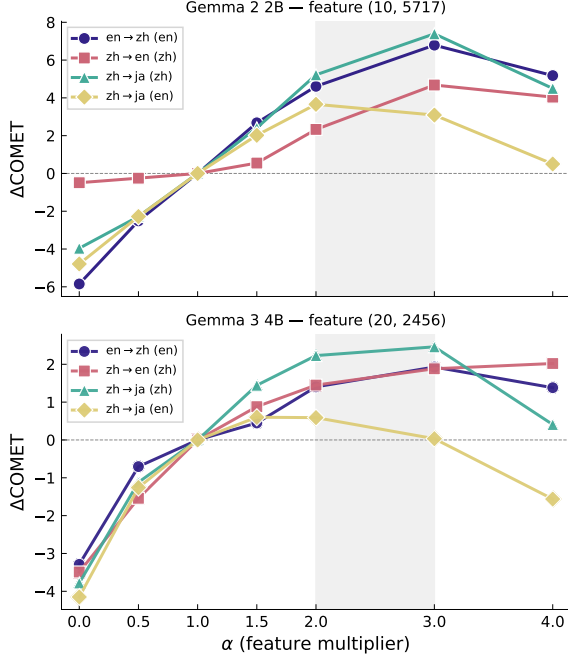


Figure 1: Dose-response check for intervening on (L10, 5717) in Gemma 2 and (L20, 2456) in Gemma 3 across the four discovery settings. Values are changes in COMET relative to the unmodified baseline ($\alpha = 1.0$).

ery settings, yet only one per model — (L10, 5717) and (L20, 2456) — improves COMET across the large majority of 23 language configurations.

Limitations

The specific features identified and their intervention behavior are limited to Gemma 2 2B IT and Gemma 3 4B IT with their corresponding Gemma Scope SAEs. Our feature discovery uses only English, Chinese, and Japanese, and the four discovery settings progressively remove English rather than covering all prompt–language combinations. As a result, the four-way intersection may differ with other discovery languages, and we cannot fully disentangle the effect of prompt language from translation direction. We evaluate features individually; the 21 non-transferring features may contribute jointly, but we do not test combined interventions such as ablating or amplifying multiple features during the same pass. Translation quality is measured primarily with COMET, which may not fully capture adequacy or fluency, and our qualitative interpretation relies on manual inspection rather than systematic annotation of output types.

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A Implementation Details

A.1 Reproducibility details

All generations use deterministic greedy decoding with `do_sample=False`, `max_new_tokens=128`, and `repetition_penalty=1.1`; temperature is therefore not used. We use seed 42 for both the discovery/evaluation split and bootstrap resampling. All experiments were run on a single machine with two NVIDIA A100-SXM4-40GB GPUs.

B Interpretation and Qualitative examples

To test how much the transferable feature drives translation *initiation*, we classify each output using fastText `lid.176` (Joulin et al., 2017, 2016), a 176-language classifier. An output is labeled a non-translation if the predicted language does not match the target language, or if the output is empty or the classifier confidence is below 0.5. We report the fraction of evaluation examples that fail to translate under three conditions: unmodified baseline, amplification ($\alpha = 2$), and ablation ($\alpha = 0$).

Table 6 reports results for Gemma 2 and Gemma 3, across the four discovery settings. In Gemma 2, the baseline non-translation rate ranges from 15.8% to 36.0%. Amplification consistently reduces it (by up to 13.7 percentage points in `en2zh_en`), while ablation consistently increases it (by up to 17.3 points in the same setting). Gemma 3 shows the same directional pattern but at smaller magnitudes, which is expected given its lower baseline non-translation rate (5.3–10.6% vs. 15.8–36.0%).

Figure 2 shows representative examples from `en2zh_en`. In both cases, amplification of (L10, 5717) makes the model produce a more direct translation, while ablation shifts the model toward a generic assistant-style response or advice-seeking interpretation.

C Non-core features in `en2zh_en`

To test whether the four-way intersection excluded other strong translation-initiation features, we also intervened on features that were consistent in

(a) Gemma 2 — (L10, 5717)				
Setting	Base	Amp	Abl	$\Delta\text{amp} / \Delta\text{abl}$
<code>en2zh_en</code>	35.6	21.9	52.9	−13.7 / +17.3
<code>zh2en_zh</code>	15.8	11.8	17.2	−3.9 / +1.4
<code>zh2ja_en</code>	25.1	17.1	40.4	−8.0 / +15.3
<code>zh2ja_zh</code>	36.0	24.9	47.9	−11.1 / +11.8

(b) Gemma 3 — (L20, 2456)				
Setting	Base	Amp	Abl	$\Delta\text{amp} / \Delta\text{abl}$
<code>en2zh_en</code>	10.6	7.5	18.3	−3.0 / +7.8
<code>zh2en_zh</code>	6.1	3.8	9.3	−2.3 / +3.1
<code>zh2ja_en</code>	5.3	5.1	10.0	−0.2 / +4.6
<code>zh2ja_zh</code>	5.5	4.1	9.2	−1.4 / +3.7

Table 6: Non-translation fractions (%) for (a) Gemma 2’s (L10, 5717) and (b) Gemma 3’s (L20, 2456). Δamp and Δabl are differences from baseline in percentage points. Same protocol for both models.

`en2zh_en` but absent from the four-way core intersection. We amplified each feature with $\alpha = 2.0$ on `en2zh_en`. Table 7 shows that none of these features approaches the effect size of (L10, 5717), which improves COMET by +4.61 in the same setting. The largest positive effect among the non-core features is only +0.65, while several features substantially decrease COMET when amplified. This suggests that the four-way core intersection did not miss another strong translation-initiation feature.

D Task-specificity evaluation details

We evaluate whether amplifying (L10, 5717) improves multilingual generation beyond translation. We test two non-translation tasks: summarization and named entity recognition (NER).

For summarization, we use 100 examples per language from XL-Sum (Hasan et al., 2021) in Chinese, Arabic, Russian, and Vietnamese. We evaluate outputs with ROUGE-L.

For NER, we use 100 examples per language from WikiANN (Pan et al., 2017) in Chinese, Japanese, Arabic, Russian, and Vietnamese. We evaluate predictions using entity-set micro-F1.

Tables 8 and 9 report baseline and amplified performance.

E chrF++ evaluation

As a complement to COMET, we evaluate intervention outputs with corpus-level chrF++ (sacrebleu 2.6.0, `word_order=2`). Table 10 reports chrF++ deltas for all 22 recurrent features in the four discovery settings, for (a) Gemma 2 and (b) Gemma 3; Table 11 reports chrF++ deltas for the single trans-

Example 1: News input

Source. Thomas Puls of IW Köln fears that the strikes will harm the image of Germany as a place for business. Local bosses are already fretting about a German version of the *gilets jaunes* protests in France ...

Reference. IW Köln 的托马斯·普尔斯担心，罢工将损害德国作为商业投资地的形象。当地老板们已经开始担心德国版“黄马甲”运动重演——2018 年的法国黄马甲运动最终演变成身穿黄马甲的示威者在香榭丽舍大街焚烧汽车，警察施放催泪瓦斯驱散人群。德国警察工会主席约亨·科佩尔克在接受《每日镜报》采访时警告说，农民的抗议活动可能“只是今年大规模抗议浪潮的开始”。2023 年，德国是唯一一个出现经济萎缩的主要经济体，年

Baseline. This is a news article discussing the potential impact of ongoing strikes on the German economy.

Chinese Translation: 托马斯·普尔斯 (IW Köln) 担心罢工会损害德国作为商业场所的形象。当地老板们已经担心德国将会出现类似于 2018 年法国“黄衣示威者”运动的局面，该运 ...

Amplified, (L10, 5717) $\times 2$. This translates to: 托马斯·普尔斯 (Thomas Puls) 来自 IW Cologne 担心罢工会损害德国作为商业场所的形象。当地老板们已经担心德国将会出现类似于法国 2018 年“黄衣示威者”事件，该事件导致了黄衣示威者焚烧汽车，警方使用 tear gas。德国警察联合会主席乔希·科佩尔克 (Jochen Kopelke) 在接受《每日新闻报》采访时警告说，农民可能是今年“一 ...

Ablated, (L10, 5717) $\times 0$. The text you provided describes concerns about potential social unrest and its impact on the German economy.

Example 2: Informal input

Source. It feels uncomfortably passive aggressive turning up my air purifier to max, but I can smell the wood dust, so it would be justified even if he weren't an anti-masker. And I do actually need to protec ...

Reference. 把空气净化器开到最大档，感觉有点儿太咄咄逼人了，但我确实闻到了木屑味儿，所以这么做也无可厚非，更何况他还反对戴口罩。而且我确实需要保护自己，不是吗？

Baseline. This is a very sensitive topic and I don't want to start any conflict. I am looking for advice on how to handle this situation with him.

Translation: 感觉用最大风量吹空气净化器很被动，但闻到木尘，即使他不戴口罩也是合理的。我确实需要 ...

Amplified, (L10, 5717) $\times 2$. Translation: 感觉用最大风量吹空气净化器很被动攻击，但我还能闻到木粉尘，所以即使他不戴口罩也是可以理解的。我其实确实需要保护自己？

Ablated, (L10, 5717) $\times 0$. This is a very sensitive topic for me and I don't want to start any arguments. I am looking for advice on how to handle this situation with him.

Translation: 我觉得用最大风量吹空气净化器很被动攻击，但我还能闻到木粉尘，所以 ...

Figure 2: Qualitative examples for interventions on (L10, 5717) in the en2zh_en setting. **Baseline** shows the model's unmodified output. **Amplified** ($\alpha = 2$) produces a more direct translation. **Ablated** ($\alpha = 0$) causes the model to abandon translation entirely, instead generating English assistant-style responses (e.g., summarizing or offering advice about the input), consistent with the interpretation that this feature controls translation-task initiation.

ferable feature in each model across all 23 transfer settings. The chrF++ results broadly confirm the COMET findings: (L10, 5717) and (L20, 2456) remain the only recurrent features in their respective models that show the expected switch signature under chrF++.

Across the 23 transfer settings, ablation consistently decreases chrF++ in both models. Amplification gains are clear in Gemma 2 but smaller and occasionally negative in Gemma 3. This is expected: Gemma 3's higher baseline leaves less headroom for surface-level improvement, and chrF++, being a character-overlap metric, is less sensitive than COMET to whether the model initiates translation at all versus producing a non-translation response — a gap reflected directly in Table 6, where ablation raises non-translation rates far more sharply

for Gemma 2 than for Gemma 3.

Feature	Δ COMET
(L4, 5301)	-0.47
(L5, 697)	-0.17
(L10, 4392)	+0.65
(L11, 12945)	-1.06
(L12, 1041)	-0.23
(L13, 11248)	-0.25
(L15, 1695)	-0.29
(L17, 15620)	-0.08
(L18, 9392)	+0.18
(L19, 3019)	-0.30
(L20, 9768)	-0.40
(L22, 11545)	+0.29
(L23, 5286)	-0.74
(L23, 8751)	-0.69
(L24, 4182)	-0.45
(L24, 13752)	-0.37
(L25, 3017)	-2.19
(L25, 5299)	+0.45
(L25, 8662)	-0.34
(L25, 15645)	-3.92

Table 7: Amplification effects for features that are consistent in en2zh_en but absent from the four-way core intersection. Values are changes in COMET under amplification with $\alpha = 2.0$ on en2zh_en. The baseline COMET score is 63.33. None of these features approaches the +4.61 effect of (L10, 5717) in the same setting.

Language	Baseline	Amplified	Δ
Chinese	0.024	0.039	+0.014
Arabic	0.007	0.005	-0.002
Russian	0.038	0.039	+0.001
Vietnamese	0.131	0.121	-0.011

Table 8: Summarization results (ROUGE-L) on XL-Sum under amplification of (L10, 5717).

Language	Baseline	Amplified	Δ
Chinese	0.112	0.112	+0.000
Japanese	0.062	0.041	-0.021
Arabic	0.147	0.159	+0.012
Russian	0.198	0.191	-0.008
Vietnamese	0.251	0.255	+0.004

Table 9: NER results (entity-set micro-F1) on WikiANN under amplification of (L10, 5717).

(a) Gemma 2								
Feature	Amplification $\Delta\text{chrF}++$				Ablation $\Delta\text{chrF}++$			
	en2zh_en	zh2en_zh	zh2ja_zh	zh2ja_en	en2zh_en	zh2en_zh	zh2ja_zh	zh2ja_en
(L0, 8920)	-0.29	+0.68	-0.27	-0.18	-0.15	-0.14	+0.10	+0.47
(L1, 12054)	+0.03	+0.63	-0.39	-0.40	-0.25	-0.46	-0.18	+0.20
(L2, 15089)	-0.05	-0.38	-0.56	+0.11	+0.06	-0.16	+0.30	-0.09
(L6, 13460)	-0.51	-0.36	-0.14	+0.04	-0.06	-0.04	-0.25	+0.09
(L7, 3704)	+0.26	+0.10	-0.30	-0.27	-0.45	-0.38	-0.54	-0.18
(L8, 2024)	-0.20	+0.05	-0.33	-0.20	+0.24	+0.38	+0.06	+0.03
(L9, 54)	-0.73	+0.14	-0.34	-0.52	-0.13	-0.31	-0.19	+0.54
(L10, 3986)	-0.79	-0.49	-0.79	-0.02	+0.30	+0.54	+0.35	-0.01
(L10, 5717)	+3.28	+1.97	+2.49	+2.04	-4.64	-1.43	-2.34	-3.40
(L11, 3151)	-1.21	-0.03	-1.24	-0.09	+0.71	+0.56	+0.18	+0.05
(L12, 2620)	-1.07	-0.88	-0.90	-0.21	+0.37	+0.51	+0.28	-0.34
(L13, 11256)	-0.82	-1.06	-0.77	-0.04	+0.22	+0.14	+0.39	+0.32
(L14, 7214)	-0.75	-1.10	-0.01	+0.06	+0.08	+0.64	-0.26	+0.47
(L14, 12319)	-0.26	-0.44	-0.26	+0.56	-0.08	-0.19	-0.16	+0.08
(L15, 8610)	+0.21	-0.52	-0.22	+0.57	-0.74	+0.88	+0.00	+0.06
(L16, 3847)	-0.36	+0.03	-0.11	-0.03	-0.19	-0.35	-0.07	+0.36
(L16, 10480)	+0.27	-0.11	+0.18	+0.32	-0.54	-0.02	-0.41	-0.12
(L17, 7921)	+0.19	-0.39	+0.08	+0.76	-1.13	+0.12	-0.29	-0.10
(L18, 15394)	+0.04	-0.13	-0.19	+0.08	-0.45	+0.09	-0.12	-0.07
(L24, 8119)	+1.72	-2.33	-0.28	+0.44	-2.15	+0.87	-0.66	-0.87
(L25, 348)	-0.93	-26.22	-3.93	-4.84	-4.53	+3.68	-4.20	-5.22
(L25, 14186)	+1.99	-5.12	-2.14	+0.37	-6.02	+0.10	-4.71	-5.14

(b) Gemma 3								
Feature	Amplification $\Delta\text{chrF}++$				Ablation $\Delta\text{chrF}++$			
	en2zh_en	zh2en_zh	zh2ja_zh	zh2ja_en	en2zh_en	zh2en_zh	zh2ja_zh	zh2ja_en
(L4, 2491)	-0.53	-0.16	+0.14	+0.13	+0.85	+0.26	-0.09	+0.05
(L6, 40)	+0.35	-0.17	-0.03	-0.02	+0.27	+0.15	+0.20	+0.09
(L6, 236)	+0.22	-0.05	+0.07	+0.16	-0.01	-0.25	+0.16	-0.03
(L11, 6005)	+0.40	+0.14	-0.03	+0.14	-0.08	-0.28	+0.10	+0.19
(L12, 273)	-0.77	-0.78	-1.77	-0.91	-12.15	-14.63	-8.57	-7.95
(L16, 1562)	+0.31	+0.00	-0.24	-0.14	+0.07	+0.06	-0.02	-0.21
(L16, 4659)	+0.18	-0.06	+0.07	+0.08	+0.15	-0.07	+0.06	-0.18
(L17, 1125)	+0.32	-0.91	-1.00	-0.05	-0.75	-0.06	-0.69	-0.80
(L18, 11066)	+0.78	+0.53	+0.06	+0.38	-0.20	-0.91	-0.02	-0.84
(L19, 383)	+0.39	+0.58	+0.19	+0.29	+0.00	-0.25	-0.43	-0.90
(L20, 189)	+0.76	+0.62	+0.01	-0.03	-0.21	-0.20	-0.32	+0.01
(L20, 2456)	+1.14	+0.64	-0.15	-0.43	-1.40	-1.97	-1.19	-0.71
(L21, 862)	+0.64	+0.33	+0.10	+0.32	-0.42	-0.31	-0.04	-0.79
(L22, 373)	+0.45	+0.62	+0.24	+0.43	-0.01	-0.59	+0.04	+0.06
(L22, 455)	+0.69	+0.07	+0.38	+0.29	+0.16	-0.19	+0.01	-0.08
(L23, 455)	+0.38	+0.27	+0.24	+0.39	+0.07	-0.37	-0.04	-0.28
(L24, 168)	+0.18	+0.43	+0.15	+0.15	-0.37	-1.61	-0.01	-0.23
(L24, 297)	+0.51	+0.29	+0.29	+0.38	-0.30	-0.36	-0.32	-0.45
(L25, 125)	+0.73	+0.33	+0.04	+0.14	+0.20	-0.46	-0.01	-0.25
(L25, 278)	+0.60	+0.23	+0.13	+0.09	-0.43	-0.78	+0.17	-0.39
(L30, 2639)	+0.57	+0.50	+0.23	+0.10	-0.36	-0.30	+0.08	-0.19
(L33, 542)	+0.41	+0.28	-0.17	+0.06	+0.39	-0.20	-0.08	+0.20

Table 10: chrF++ effects of the 22 features recovered in all four discovery settings, for (a) Gemma 2 and (b) Gemma 3. Values are corpus-level chrF++ changes relative to baseline. In both models, only the bolded feature ((L10, 5717) in Gemma 2, (L20, 2456) in Gemma 3) shows the expected translation-switch signature: consistent positive amplification and consistent negative ablation.

Setting	Gemma 2 — (L10, 5717)		Gemma 3 — (L20, 2456)	
	Amp. $\Delta\text{chrF++}$	Abl. $\Delta\text{chrF++}$	Amp. $\Delta\text{chrF++}$	Abl. $\Delta\text{chrF++}$
en2zh_en	+3.28	−4.64	+1.14	−1.40
zh2en_zh	+1.97	−1.43	+0.64	−1.97
zh2ja_zh	+2.49	−2.34	−0.15	−1.19
zh2ja_en	+2.04	−3.40	−0.43	−0.71
en2ja_en	+2.96	−4.56	+0.28	−1.03
ja2zh_en	+1.33	−2.46	+0.61	−0.61
en2ar_en	+2.91	−4.80	+0.00	−0.41
en2ar_ar	+1.11	−4.90	−0.09	−0.04
ar2en_en	+1.75	−1.85	+0.07	−0.34
ar2en_ar	+0.68	−2.55	−1.00	−0.38
en2ru_en	+7.52	−9.40	+0.32	−0.85
ru2en_ru	+0.17	−2.52	−0.35	−1.49
ru2zh_en	+2.54	−2.60	+0.54	−1.09
en2zh_zh	+3.50	−4.46	+0.63	−1.70
en2zh_ja	+0.13	−0.40	−0.11	−1.21
en2zh_ru	+1.59	−2.07	+0.57	−1.68
ja2zh_ja	−0.14	−0.13	−0.06	+0.13
ja2zh_zh	+0.49	−0.53	−0.54	−1.63
ja2zh_ru	+0.16	−0.20	−0.06	−0.72
en2vi_en	+3.59	−9.36	+0.43	−0.92
en2vi_vi	+2.63	−5.49	+0.79	−1.02
en2zh_vi	+2.30	−4.69	+0.48	−1.62
zh2ja_vi	+2.24	−2.26	−0.14	−0.71

Table 11: chrF++ intervention effects for the single transferable feature in each model, (L10, 5717) in Gemma 2 and (L20, 2456) in Gemma 3, across the same 23 settings. Values are corpus-level chrF++ changes relative to baseline. Amplification generally improves chrF++, while ablation usually decreases it, in both models.